Spark Lens: Product Blueprint & Project Structure

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Executive Summary

Spark Lens is an AI-powered repository intelligence and security platform. Unlike disjointed AI dev tools, Spark Lens provides a singular, cohesive workflow: it transforms raw repository data into actionable, verifiable, and remediated security insights. By connecting directly to GitHub, it acts as an autonomous security engineer that not only flags issues but proves they exist and helps fix them.

The Core Workflow (The Golden Path)

To avoid feature bloat, every capability in Spark Lens serves this linear progression:

1. **Connect & Ingest:**
    - **Action:** Developer authorizes Spark Lens via GitHub App/OAuth.
    - **Result:** Platform ingests repository context, dependency trees, and recent commits.
2. **Analyze & Explain:**
    - **Action:** AI and static analysis engines scan for code quality issues, hardcoded secrets, and logical vulnerabilities.
    - **Result:** Developer receives a human-readable explanation of *what* the code is doing and *where* it is vulnerable.
3. **Verify (The Sandbox):**
    - **Action:** Spark Lens attempts to generate a Proof of Concept (PoC) or test case in an isolated, controlled container environment.
    - **Result:** Eliminates false positives by proving the vulnerability is reproducible.
4. **Prioritize:**
    - **Action:** Issues are ranked based on exploitability (from the Verify stage) and business impact.

    - **Result:** A focused dashboard highlighting critical, verified threats first.
5. **Fix & Remediate:**
    - **Action:** AI generates contextual, tested patch recommendations (Pull Requests).
    - **Result:** 1-click PR generation to patch the vulnerability.
6. **Validate:**
    - **Action:** The system re-runs the analysis and verification tests on the patched branch.
    - **Result:** Confirmation that the fix is successful without breaking existing functionality.

High-Level Project Structure

1. Frontend (User Interface)

- **Tech Stack:** React / Next.js, Tailwind CSS
- **Core Views:**
    - **Dashboard:** High-level metrics, security score, and pending critical actions.
    - **Repo Lens:** Deep dive into a specific repository's analysis pipeline.
    - **Vulnerability Studio:** Interactive view for the Explain -> Verify -> Fix stages.

2. Backend & Integrations

- **Tech Stack:** Node.js/Express or Python/FastAPI, PostgreSQL
- **Core Modules:**
    - **GitHub Integration Manager:** Handles OAuth, Webhooks (triggering on PRs/commits), and API rate limits.
    - **Workflow Orchestrator:** Manages the state machine of the core workflow (Connect -> Validate).

3. AI & Analysis Engine

- **Tech Stack:** LangChain/LlamaIndex, OpenAI/Anthropic APIs, Static Analyzers (e.g., Semgrep).
- **Core Modules:**
    - **Context Engine:** Chunks and embeds repository code to give the LLM accurate context.
    - **Vulnerability Explainer:** Translates complex flaws into simple summaries.
    - **Remediation Generator:** Writes the actual patch code.

4. Execution Environment (The "Verify" Engine)

- **Tech Stack:** Docker, Kubernetes (for isolated sandboxing).
- **Core Modules:**
    - **Ephemeral Containers:** Spins up secure sandboxes to safely run AI-generated test cases and verify exploits.

Key Differentiator

Spark Lens's moat is the **Verify** phase. While other tools stop at "Analyze" (creating noise and false positives), Spark Lens proves the issue exists in a sandbox before asking a developer to spend time fixing it.